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CLAMPING DEVICE, ELECTRONIC DEVICE ASSEMBLY, AND BICYCLE

Abstract

Fastening unit configured to fix a distance between a first adjusting portion and a second adjusting portion in a circumferential direction of a handlebar is provided. A first engaging portion and a second engaging portion are located between a first engaged portion and a second engaged portion in a tangential direction of an outer circumferential surface of the handlebar. The fastening unit allows adjustment of the distance between the first adjusting portion and the second adjusting portion. A distance between the first engaging portion and the second engaging portion increases due to a decrease in the distance between the first adjusting portion and the second adjusting portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority from Japanese application JP2024-036254 filed on Mar. 8, 2024, the content of which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to a clamping device, an electronic device assembly, and a bicycle.

2. Description of the Related Art

[0003] A handlebar of a bicycle may be equipped with an electronic device that displays vehicle speed, traveled distance, time, and the like. In JP 2544082 Y2, such an electronic device is detachable from a clamping device (a component attaching device in JP 2544082 Y2) attached to a handlebar. As illustrated in JP 2015-085919 A, an electronic device attached to a handlebar of an electrically assisted bicycle may display an assist mode, a remaining battery charge amount, and the like and have a switch (button) for switching the assist mode.

[0004] Other than the electronic device described above that displays vehicle speed and the like, various parts and devices may be attached to a handlebar of a bicycle. For example, a clamping device for supporting a mobile phone, a lighting device, and the like may be attached to the handlebar. Therefore, flexibility in positioning of the clamping device is limited, so that it may be difficult to attach the electronic device in an optimal position. For example, a button may be located far away from a grip.

[0005] An object of the present disclosure is to provide a clamping device that enables position adjustment of an electronic device and that can facilitate easily adjusting a position of the electronic device.

SUMMARY OF THE INVENTION

[0006] (1) An example of a clamping device proposed in the present disclosure is a clamping device for attaching an electronic device including a first engaged portion and a second engaged portion to a handlebar. The clamping device includes a first clamping member having a first inner circumferential surface extending along a portion of an outer circumferential surface of the handlebar, a first engaging portion engageable with the first engaged portion to allow movement of the first engaged portion in a longitudinal direction of the handlebar, and a first adjusting portion, a second clamping member having a second inner circumferential surface extending along another portion of the outer circumferential surface of the handlebar, a second engaging portion engageable with the second engaged portion to allow movement of the second engaged portion in the longitudinal direction of the handlebar, and a second adjusting portion, and fastening unit configured to fix a distance between the first adjusting portion and the second adjusting portion in a circumferential direction of the handlebar. The first engaging portion and the second engaging portion are located between the first engaged portion and the second engaged portion in a tangential direction of the outer circumferential surface of the handlebar. The fastening unit allows adjustment of the distance between the first adjusting portion and the second adjusting portion. A distance between the first engaging portion and the second engaging portion increases due to a decrease in the distance between the first adjusting portion and the second adjusting portion.

[0007] According to the clamping device, both of a relative position between the clamping device and the handlebar and a relative position between the clamping device and the electronic device can be adjusted or fixed by operating the fastening unit. Therefore, position adjustment thereof can be facilitated.

[0008] (2) The clamping device according to (1) may further include a connecting shaft portion.

The first clamping member and the second clamping member respectively have a first connected portion and a second connected portion located between the first engaging portion and the second engaging portion in the tangential direction of the outer circumferential surface of the handlebar. The connecting shaft portion connects the first connected portion and the second connected portion to allow an increase and decrease in the distance between the first engaging portion and the second engaging portion. Such a structure makes it possible to prevent separation of the two clamping members. As a result, the clamping device is easy to handle.

[0009] (3) In the clamping device according to (2), the connecting shaft portion protrudes from the first connected portion in the longitudinal direction of the handlebar. A connecting hole into which the connecting shaft portion is fitted may be formed in the second connected portion. The connecting hole may be larger than the connecting shaft portion in the tangential direction of the outer circumferential surface of the handlebar.

[0010] (4) In the clamping device according to (3), a distance between the connecting shaft portion and the outer circumferential surface of the handlebar may be smaller than a distance between the outer circumferential surface of the handlebar and a first plane passing through the first engaging portion and the second engaging portion, the first plane extending along a first tangent line to the outer circumferential surface of the handlebar. Such an arrangement of the connecting shaft portion allows a size of the connecting hole in the tangential direction of the outer circumferential surface of the handlebar to be decreased. As a result, a decrease in strength of the second clamping member can be prevented.

[0011] (5) In the clamping device according to any one of (1) to (4), the first adjusting portion, a contact surface between the first inner circumferential surface and the outer circumferential surface of the handlebar, and the first engaging portion respectively function as a point of effort, a fulcrum point, and a point of load. The second adjusting portion, a contact surface between the second inner circumferential surface and the outer circumferential surface of the handlebar, and the second engaging portion respectively function as a point of effort, a fulcrum point, and a point of load.

[0012] (6) In the clamping device according to any one of (1) to (5), the distance between the outer circumferential surface of the handlebar and the first plane and passing through the first engaging portion and the second engaging portion may be greater than a distance between the outer circumferential surface of the handlebar and a second plane passing through the first adjusting portion and the second adjusting portion, the first plane extending along the first tangent line to the outer circumferential surface of the handlebar, and the second plane extending along a second tangent line to the outer circumferential surface of the handlebar. Accordingly, when positions of the two adjusting portions are fixed by the fastening unit, it is possible to prevent an excessive force from acting between the first engaging portion of the first clamping member and the first engaged portion of the electronic device, and prevent an excessive force from acting between the second engaging portion of the second clamping member and the second engaged portion of the electronic device.

[0013] (7) In the clamping device according to any one of (1) to (6), the distance between the first engaging portion and the second engaging portion in a first direction passing through the first engaging portion and the second engaging portion may be greater than the distance between the first adjusting portion and the second adjusting portion in a second direction passing through the first adjusting portion and the second adjusting portion, the first direction extending along the first tangent line to the outer circumferential surface of the handlebar, and the second direction extending along the second tangent line to the outer circumferential surface of the handlebar.

Accordingly, the distance between the first engaging portion and the second engaging portion increases, thereby improving stability of the clamping device in supporting the electronic device. Since the distance between the first adjusting portion and the second adjusting portion decreases, the fastening unit can be prevented from standing out by the handlebar.

[0014] (8) In the clamping device according to any one of (1) to (7), when an angular position of an

intermediate position between the first engaging portion and the second engaging portion is set to 0 degrees in the circumferential direction of the handlebar, at least one of the first adjusting portion and the second adjusting portion may be located in a range of 90 degrees to 270 degrees. Accordingly, it is possible to prevent a decrease in strength of the first clamping member due to an excessive decrease in a distance between the first adjusting portion and the first engaging portion. It is possible to prevent a decrease in strength of the second clamping member due to an excessive decrease in a distance between the second adjusting portion and the second engaging portion.

[0015] (9) In the clamping device according to any one of (1) to (8), when an angular position of an intermediate position between the first engaging portion and the second engaging portion is set to 0 degrees in the circumferential direction of the handlebar, an intermediate position between the first adjusting portion and the second adjusting portion may be located in a range of 135 degrees to 225 degrees. Accordingly, the first clamping member and the second clamping member can have sizes similar to each other. As a result, a difference in stress acting on each of the two clamping members can be reduced.

[0016] (10) In the clamping device according to any one of (1) to (9), the first plane passing through the first engaging portion and the second engaging portion may be substantially parallel to a direction in which the first adjusting portion and the second adjusting portion face each other, the first plane extending along the first tangent line to the outer circumferential surface of the handlebar. Accordingly, the fastening unit can be operated from a front side or a rear side of the clamping device when the clamping device is attached to the handlebar. As a result, position adjustment of the two adjusting portions by the fastening unit can be facilitated.

[0017] (11) In the clamping device according to any one of (1) to (9), the first plane passing through the first engaging portion and the second engaging portion may intersect with a direction in which the first adjusting portion and the second adjusting portion face each other, the first plane extending along the first tangent line to the outer circumferential surface of the handlebar.

[0018] (12) An example of an electronic device assembly proposed in the present disclosure includes the clamping device according to any one of (1) to (11) and the electronic device.

[0019] (13) An example of a bicycle proposed in the present disclosure includes the clamping device according to any one of (1) to (11), the electronic device, and the handlebar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a side view of a bicycle proposed in the present disclosure;

[0021] FIG. 2 is a perspective view of an electronic device assembly including a clamping device and an electronic device proposed in the present disclosure;

[0022] FIG. 3 is a perspective view of the electronic device assembly;

[0023] FIG. 4 is a perspective view of the clamping device;

[0024] FIG. 5 is an exploded perspective view of the clamping device;

[0025] FIG. 6A is a side view of the clamping device;

[0026] FIG. 6B is an enlarged view of an engaging portion of the clamping device;

[0027] FIG. 7 is a perspective view illustrating another example of the clamping device; and

[0028] FIG. 8 is a side view of the clamping device illustrated in FIG. 7.

DETAILED DESCRIPTION OF THE INVENTION

[0029] The present disclosure is to be considered as an exemplification of the invention, and is not intended to limit the invention to the specific embodiments illustrated by the figures or description below. The present invention will now be described by referencing the appended figures representing embodiments.

[0030] The terminology used herein is for the purpose of describing particular embodiments only

and is not intended to be limiting of the invention. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well as the singular forms, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, “includes”, and/or “including”, when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0031] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one having ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0032] In describing the invention, it will be understood that a number of technologies are disclosed. Each of these has individual benefit and each can also be used in conjunction with one or more, or in some cases all, of the other disclosed technologies. Accordingly, for the sake of clarity, this description will refrain from repeating every possible combination of the individual technologies in an unnecessary fashion. Nevertheless, the specification and claims should be read with the understanding that such combinations are entirely within the scope of the invention and the claims.

[0033] In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be evident, however, to one skilled in the art that the present invention may be practiced without these specific details.

[0034] A clamping device and a bicycle proposed in the present disclosure are described below. FIG. 1 is a side view illustrating an example of the bicycle proposed in the present disclosure. FIGS. 2 to 6B are diagrams illustrating an example of the clamping device proposed in the present disclosure.

[0035] In the following description, a Y1 direction and a Y2 direction illustrated in FIG. 2 or the like will be respectively referred to as a front side and a rear side. An X1 direction and an X2 direction illustrated in the drawing are respectively referred to as a right side and a left side, and a Z1 direction and a Z2 direction are respectively referred to as an upper side and a lower side. A Y1-Y2 direction is referred to as a front-rear direction, and a Z1-Z2 direction is referred to as an up-down direction. An X1-X2 direction is referred to as a left-right direction, or a longitudinal direction of a handlebar described below. Since the clamping device may be attached obliquely to a handlebar of the bicycle, the front-rear direction (Y1-Y2 direction), the left-right direction (X1-X2 direction), and the up-down direction (Z1-Z2 direction) illustrated in FIG. 2 or the like do not have to completely match the front-rear direction, the left-right direction, and the up-down direction of the bicycle.

[Bicycle]

[0036] As illustrated in FIG. 1, a bicycle 10 includes a body frame 1, a front wheel 2, a rear wheel 3, a handlebar 4, a saddle 5, a crank 6, and a pedal 7. The bicycle 10 is an electrically assisted bicycle and may include a drive unit 8 and a battery 9 that supplies power to the drive unit 8.

[0037] As illustrated in FIG. 1, the body frame 1 has a head pipe 11 at a front end thereof. The head pipe 11 rotatably supports a steering shaft. The handlebar 4 is fixed to an upper portion of the steering shaft. A grip 4a is provided at each of a right end and a left end of the handlebar 4. A front fork 23 that rotatably supports the front wheel 2 is fixed to a lower portion of the steering shaft.

[0038] As illustrated in FIG. 1, the body frame 1 may have, for example, a top frame 12, a down frame 13, a seat tube 14, a seat stay 15, a chain stay 16, and a bracket 17. The top frame 12 extends

obliquely rearward and downward from the head pipe **11** and is connected at a rear end thereof to the seat tube **14**. The down frame **13** extends obliquely rearward and downward from the head pipe **11** and is disposed below the top frame **12**. A rear end of the down frame **13** is connected to the bracket **17**.

[0039] A seat post **21** extends upward from the seat tube **14** and supports the saddle **5** disposed above it. The bracket **17** is connected to a lower end of the seat tube **14**. The seat stay **15** extends obliquely rearward and downward from the seat tube **14**. The chain stay **16** extends rearward from the bracket **17**. A lower end (rear end) of the seat stay **15** and a rear end of the chain stay **16** are connected. The rear wheel **3** is rotatably supported at the connection position between them.

[0040] The crank **6** has a crank in a width direction of a vehicle body and a crank arm **6a** that rotates around the crank shaft. The pedal **7** is attached to a tip of the crank arm **6a**. A chain **24** is stretched between a sprocket provided on the crank shaft and a sprocket provided on an axle of the rear wheel **3**.

[0041] The drive unit **8** includes an electric motor and a deceleration mechanism, and is held by the bracket **17**. The drive unit **8** can transmit torque to the crank **6** and the sprocket. Therefore, a force acting on the pedal **7** (force from a rider stepping on the pedal **7**) and torque output by the drive unit **8** are transmitted to the rear wheel **3** via the chain **24**.

[0042] The bicycle proposed in the present disclosure does not necessarily include the drive unit **8** described herein. A structure of the body frame **1** is not necessarily limited to the example illustrated in FIG. **1**. For example, the body frame **1** does not necessarily have the top frame **12**.

[0043] As illustrated in FIG. **2**, a clamping device **100** is attached to the handlebar **4**. An electronic device **70** is supported by the handlebar **4** via the clamping device **100**. As will be described below, the electronic device **70** is detachable from the clamping device **100**. The clamping device **100** and the electronic device **70** form an electronic device assembly.

[Electronic Device]

[0044] The electronic device **70** may be connected to a control device (not illustrated) or a sensor mounted on the bicycle **10** via a cable. The electronic device **70** includes a display device **71a**, and displays information received from the control device or the sensor, specifically information indicating status of the bicycle **10**, on the display device **71a**. The information displayed on the display device **71a** may be, for example, vehicle speed, traveled distance, number of pedal revolutions, and the like. The electronic device **70** may display information related to the drive unit **8**, such as an amount of remaining charge of the battery **9** or an operating mode of the drive unit **8**, as information indicating the status of the bicycle **10**. The operating mode includes, for example, a strong mode in which the torque output by the drive unit **8** is relatively large, a weak mode in which the torque is relatively small, and a medium mode in which the torque is intermediate between the strong mode and the weak mode.

[0045] The control device mounted on the bicycle **10** may control the drive unit **8** based on an output signal from various sensors. Here, the information indicating the status of the bicycle **10** described above may be transmitted from the control device to the electronic device **70**.

Alternatively, the output signal from the various sensors (for example, a speed sensor or a pedaling force sensor) mounted on the bicycle **10** may be transmitted directly to the electronic device **70**. Here, based on the output signal of the sensor, the electronic device **70** may output a signal for controlling the electric motor of the drive unit **8** (for example, a current command value) to a drive circuit that supplies power to the electric motor.

[0046] As illustrated in FIG. **2**, the electronic device **70** may have a plurality of buttons **74a** and **74b**. The buttons **74a** and **74b** may be provided on a front surface of a main body **71** of the electronic device **70**.

[0047] As illustrated in FIG. **3**, the electronic device **70** includes a first engaged portion **72A** and a second engaged portion **72B** on a lower surface **71b** thereof. The engaged portions **72A** and **72B** protrude downward from the lower surface **71b** of the electronic device **70**. The two engaged

portions **72A** and **72B** are separated in the front-rear direction and face each other.

[Clamping Device]

[0048] As illustrated in FIGS. **4** and **5**, the clamping device **100** includes a first clamping member **30** and a second clamping member **40** that are assembled together.

[0049] The first clamping member **30** has a first inner circumferential surface **31a** fitted along a portion of an outer circumferential surface of the handlebar **4**. The second clamping member **40** has a second inner circumferential surface **41a** fitted along another portion of the outer circumferential surface of the handlebar **4**. The handlebar **4** may have a circular cross section. The inner circumferential surfaces **31a** and **41a** may be arc-shaped to be fitted along the outer circumferential surface of the handlebar **4**.

[0050] The inner circumferential surfaces **31a** and **41a** are disposed to face each other and interpose the outer circumferential surface of the handlebar **4** in the front-rear direction. A diameter of the inner circumferential surfaces **31a** and **41a** and a diameter of the outer circumferential surface of the handlebar **4** are not necessarily completely equal to each other. For example, a gap may be partially formed between the inner circumferential surfaces **31a** and **41a** and the outer circumferential surface of the handlebar **4**.

[0051] As illustrated in FIG. **5**, the inner circumferential surface **41a** of the second clamping member **40** may have an anti-slip sheet **41b**. Similarly, the inner circumferential surface **31a** of the first clamping member **30** may have an anti-slip sheet.

[Engaging Portion of Clamping Member]

[0052] As illustrated in FIG. **4**, the first clamping member **30** includes a first engaging portion **32** engageable with the first engaged portion **72A** of the electronic device **70** at the top of the first clamping member **30**. Similarly, the second clamping member **40** includes a second engaging portion **42** engageable with the second engaged portion **72B** of the electronic device **70** at the top of the second clamping member **40**.

[0053] As illustrated in FIG. **3**, the two engaged portions **72A** and **72B** of the electronic device **70** are formed to face each other in a tangential direction of the outer circumferential surface of the handlebar **4** (a direction of a tangent line passing through an upper end of the outer circumferential surface, the Y1-Y2 direction in FIG. **3**). The inner circumferential surfaces **31a** and **41a** of the clamping members **30** and **40** are curved to be fitted along the outer circumferential surface of the handlebar **4**. Therefore, “tangential direction of the outer circumferential surface of the handlebar **4**” is also a tangential direction of a cylindrical surface formed by the inner circumferential surfaces **31a** and **41a**.

[0054] As illustrated in FIG. **6A**, the engaging portions **32** and **42** of the clamping members **30** and **40** are disposed between the engaged portions **72A** and **72B** of the electronic device **70** in the tangential direction of the outer circumferential surface of the handlebar **4**, and respectively engage with the engaged portions **72A** and **72B**. Accordingly, relative movement between the engaging portions **32** and **42** and the engaged portions **72A** and **72B** in the tangential direction of the outer circumferential surface of the handlebar **4** is restricted. Meanwhile, relative movement between the engaged portions **72A** and **72B** and the engaging portions **32** and **42** in the longitudinal direction of the handlebar **4** (the X1-X2 direction) is permitted.

[0055] As illustrated in FIG. **6B**, the engaged portions **72A** and **72B** of the electronic device **70** may be substantially L-shaped. The engaging portions **32** and **42** of the clamping device **100** may also be substantially L-shaped. Protrusion portions **32a** and **42a** of the engaging portions **32** and **42** may be fitted into recess portions **72c** formed in the engaged portions **72A** and **72B**. Accordingly, relative movement between the engaged portions **72A** and **72B** and the engaging portions **32** and **42** in the longitudinal direction of the handlebar **4** is allowed. On the other hand, relative movement between the same portions in the circumferential and tangential directions of the outer circumferential surface of the handlebar **4** is restricted.

[Adjusting Portion of Clamping Member]

[0056] As illustrated in FIG. 6A, the first clamping member **30** includes a first adjusting portion **33** at a bottom portion thereof. The second clamping member **40** includes a second adjusting portion **43** at a bottom portion thereof. The two adjusting portions **33** and **43** face each other in the tangential direction of the outer circumferential surface of the handlebar **4** (a direction of the tangent line passing through the lower end of the outer circumferential surface, the Y1-Y2 direction in FIG. 6A).

[0057] The clamping device **100** includes a fastening unit which fixes a distance W2 between the adjusting portions **33** and **43** (see FIG. 6A). The clamping device **100** may include, for example, a bolt **51** and a nut **52** as fastening unit. The nut **52** is held by the second adjusting portion **43**. A hole **33c** into which the bolt **51** is inserted is formed on the first adjusting portion **33** (see FIG. 4). The bolt **51** is inserted into the hole **33c** from a rear side of the first adjusting portion **33** (a side opposite to the second adjusting portion **43**) and is fastened to the nut **52**. As a result, the distance W2 between the adjusting portions **33** and **43** is fixed.

[0058] When the second clamping member **40** is molded from resin, the nut **52** may be held inside the second adjusting portion **43** by insert molding. Alternatively, an accommodating recess portion may be formed in the second adjusting portion **43**, and the nut **52** may be inserted into the accommodating recess portion.

[Movement of Clamping Member Due to Operation of Fastening Unit]

[0059] When the bolt **51** is tightened, the distance W2 between the adjusting portions **33** and **43** (see FIG. 6A) decreases. Conversely, when the bolt **51** is loosened, the distance W2 between the adjusting portions **33** and **43** is allowed to increase. A distance W1 between the first engaging portion **32** and the second engaging portion **42** (see FIG. 6A) in the tangential direction of the outer circumferential surface of the handlebar **4** (a direction of the tangent line passing through the upper end of the outer circumferential surface) is allowed to be decreased or increased.

[0060] As illustrated in FIG. 6A, when the inner circumferential surfaces **31a** and **41a** interpose the handlebar **4** therebetween, the distance W1 between the first engaging portion **32** and the second engaging portion **42** increases due to a decrease in the distance W2 between the adjusting portions **33** and **43**. As a result, the first engaging portion **32** and the second engaging portion **42** are respectively pressed against the first engaged portion **72A** and the second engaged portion **72B** of the electronic device **70**. A contact force (friction force) between the engaging portions **32** and **42** and the engaged portions **72A** and **72B** becomes large, and the relative position of the clamping device **100** and the electronic device **70** in the longitudinal direction of the handlebar **4** is fixed. When the distance W2 of the adjusting portions **33** and **43** decreases, a contact force between the inner circumferential surfaces **31a** and **41a** and the outer circumferential surface of the handlebar **4** increases. Therefore, the position of the clamping device **100** on the handlebar **4** is also fixed.

[0061] Conversely, an increase in the distance W2 between the adjusting portions **33** and **43** allows the distance W1 between the first engaging portion **32** and the second engaging portion **42** to be decreased. Accordingly, the contact force between the first engaging portion **32** and the first engaged portion **72A** and the contact force between the second engaging portion **42** and the second engaged portion **72B** are weakened, thereby allowing the relative position of the clamping device **100** and the electronic device **70** to change in the longitudinal direction of the handlebar **4**. In addition, the contact force between the inner circumferential surfaces **31a** and **41a** and the outer circumferential surface of the handlebar **4** is reduced. Accordingly, position adjustment of the clamping device **100** on the handlebar **4** is possible.

[0062] By using the clamping device **100** as such, it is possible to adjust both of the relative position of the clamping device **100** and the electronic device **70**, and the position of the clamping device **100** on the handlebar **4** simply by operating a single fastening unit (the bolt **51**).

[0063] The first clamping member **30** and the second clamping member **40** are connected to each other only via a connecting shaft portion **53** described below, and are not connected to each other in any other portions. Therefore, when the distance W2 of the adjusting portions **33** and **43** is

decreased or increased, the distance W1 of the engaging portions **32** and **42** is increased or decreased accordingly.

[Size of Clamping Device]

[0064] As illustrated in FIG. 3, the engaged portions **72A** and **72B** of the electronic device **70** extend in the longitudinal direction of the handlebar **4** (the X1-X2 direction). A size of the engaged portions **72A** and **72B** in the longitudinal direction of the handlebar **4** (X1-X2 direction) is different from a size of the clamping members **30** and **40** in the same direction. More specifically, the size of the engaged portions **72A** and **72B** of the electronic device **70** is larger than the size of the clamping members **30** and **40** in the same direction. The engaging portions **32** and **42** are relatively slidable along the engaged portions **72A** and **72B**. The relative position of the electronic device **70** and the clamping device **100** can be adjusted within a range of the engaged portions **72A** and **72B**.

[0065] As illustrated in FIG. 3, the electronic device **70** includes an operation unit **73** on the left side. The operation unit **73** extends downward from the main body **71**. The operation unit **73** is provided with a switch **73a**. The switch **73a** may be, for example, a switch for switching the operating mode of the drive unit **8** (strong mode, medium mode, weak mode). The switch **73a** may also be an on/off switch for a lighting device.

[0066] The engaged portions **72A** and **72B** are formed on the lower surface **71b** of the main body **71** closer to the left side. That is, the engaged portions **72A** and **72B** are formed closer to the operation unit **73**. Centers of the engaged portions **72A** and **72B** in the left-right direction (the longitudinal direction of the handlebar **4**) are located to the left of a center of the main body **71** of the electronic device **70** in the left-right direction. Such engaged portions **72A** and **72B** can prevent the clamping device **100** from supporting the right portion of the main body **71** (a position far from the operation unit **73**). As a result, for example, when the operation unit **73** is pressed, support of the electronic device **70** can be prevented from becoming unstable.

[0067] When the bolt **51** is tightened, a force is applied to the first adjusting portion **33** from the fastening unit (the bolt **51**), and the first adjusting portion **33** is moved forward (in the Y1 direction). Here, the inner circumferential surface **31a** is pressed against the outer circumferential surface of the handlebar **4**. The first engaging portion **32** is moved in an opposite direction to the first adjusting portion **33**, that is, rearward (in the Y2 direction), and is pressed against the first engaged portion **72A** of the electronic device **70**. In other words, in these movements, the first adjusting portion **33**, the contact surface between the inner circumferential surface **31a** and the outer circumferential surface of the handlebar **4**, and the first engaging portion **32** respectively function as a point of effort, a fulcrum point, and a point of load.

[0068] When the bolt **51** is tightened, a force is applied to the second adjusting portion **43** from the fastening unit (the nut **52**), and the second adjusting portion **43** is moved rearward (in the Y2 direction). Here, the inner circumferential surface **41a** is pressed against the outer circumferential surface of the handlebar **4**. The second engaging portion **42** is moved in an opposite direction to the second adjusting portion **43**, that is, forward (in the Y1 direction), and is pressed against the second engaged portion **72B** of the electronic device **70**. In other words, in these movements, the second adjusting portion **43**, the contact surface between the inner circumferential surface **41a** and the outer circumferential surface of the handlebar **4**, and the second engaging portion **42** respectively function as a point of effort, a fulcrum point, and a point of load.

[Details of Engagement Structure]

[0069] As illustrated in FIG. 6B, a rear surface **32b** (the surface facing the Y2 direction) of a base of the first engaging portion **32** is pressed against a tip of a protrusion portion **72e** of the first engaged portion **72A**. A base of a lower surface **32c** of the protrusion portion **32a** of the first engaging portion **32** is also pressed against the tip of the protrusion portion **72e** of the first engaged portion **72A**. An upper surface **32d** of the first engaging portion **32** (an upper surface of the first clamping member **30**) is pressed against the lower surface **71b** of the electronic device **70**.

[0070] Similarly, a front surface **42b** (a surface facing the Y1 direction) of a base of the second

engaging portion **42** is pressed against a tip of the protrusion portion **72e** of the second engaged portion **72B**. A base of a lower surface **42c** of the protrusion portion **42a** of the second engaging portion **42** is also pressed against the tip of the protrusion portion **72e** of the second engaged portion **72B**. An upper surface **42d** of the second engaging portion **42** (an upper surface of the second clamping member **40**) is pressed against the lower surface **71b** of the electronic device **70**. [0071] With such a structure, when the distance **W2** between the adjusting portions **33** and **43** (see FIG. **6A**) decreases, the clamping members **30** and **40** are rotated along the outer circumferential surface of the handlebar **4** so that the distance **W1** between the two engaging portions **32** and **42** increases. As a result, the contact force at each of the above-described portions increases. In particular, between the engaging portions **32** and **42** and the engaged portions **72A** and **72B**, not only the contact force in the front-rear direction but also the contact force in the up-down direction increases. The contact force in the up-down direction is specifically the contact force between the upper surfaces **32d** and **42d** of the engaging portions **32** and **42** and the lower surface **71b** of the electronic device **70**, and the contact force between the lower surfaces **32c** and **42c** of the engaging portions **32** and **42** and the engaged portions **72A** and **72B**. Accordingly, the relative position of the electronic device **70** regarding the clamping device **100** is fixed in the front-rear direction, the up-down direction, and the left-right direction.

[0072] As illustrated in FIG. **6B**, the upper surfaces **32d** and **42d** of the engaging portions **32** and **42** respectively have inclined surfaces **32e** and **42e**. The inclined surfaces **32e** and **42e** are gradually lowered toward the tips of the protruding portions **32a** and **42a**.

[0073] When the distance **W2** between the adjusting portions **33** and **43** increases, the clamping members **30** and **40** are rotated along the outer circumferential surface of the handlebar **4** so that the distance **W1** between the two engaging portions **32** and **42** decreases. Therefore, the engaging portions **32** and **42** are separated from the protruding portions **72e** of the engaged portions **72A** and **72B**. Here, the inclined surfaces **32e** and **42e** face straight up (that is, the inclined surfaces **32e** and **42e** come in contact with the lower surface **71b** of the main body **71**). Accordingly, the contact force between the engaging portions **32** and **42** and the lower surface **71b** of the main body **71** decreases, making it possible to move the electronic device **70** in the left-right direction regarding the clamping device **100**. The clamping device **100** itself can also be moved in the left-right direction.

[0074] Each of the clamping members **30** and **40** may be integrally molded from resin. Here, when the bolt **51** is tightened, the clamping members **30** and **40** can be slightly elastically deformed. As a result, the engagement between the engaging portions **32** and **42** and the engaged portions **72A** and **72B** can be prevented from loosening due to vibration of the bicycle **10**, for example. Alternatively, the clamping members **30** and **40** may be made of metal.

[Connected Portion of Clamping Member and Connecting Shaft Portion]

[0075] As illustrated in FIG. **5**, the first clamping member **30** and the second clamping member **40** respectively have a first connected portion **34** and a second connected portion **44**. The connected portions **34** and **44** are located between the two engaging portions **32** and **42**.

[0076] In detail, in a plan view of the clamping device **100**, the connected portions **34** and **44** are located between the engaging portions **32** and **42**. In a side view of the clamping device **100**, the engaging portions **32** and **42** and the connected portions **34** and **44** are located in the same direction regarding the outer circumferential surface of the handlebar **4**. In detail, a plane **P1** (see FIG. **6A**) passing through the engaging portions **32** and **42** and extending along a tangent line to the outer circumferential surface of the handlebar **4** is located above the outer circumferential surface of the handlebar **4**. Similar to the engaging portions **32** and **42**, the connected portions **34** and **44** are located above the outer circumferential surface of the handlebar **4**.

[0077] As illustrated in FIGS. **4** and **5**, the two connected portions **34** and **44** are assembled together. In detail, a recess portion **44b** (see FIG. **5**) may be formed in the second connected portion **44**. The first connected portion **34** may be fitted into the recess portion **44b**. The second connected

portion **44** may have two side wall portions **44c**. The recess portion **44b** may be formed between the two side wall portions **44c**. The first connected portion **34** overlaps with the side wall portion **44c** of the second connected portion **44** in a side view of the clamping device **100**.

[0078] The clamping device **100** includes the connecting shaft portion **53** (see FIG. **4**) that connects the connected portions **34** and **44** together. The connecting shaft portion **53** is inserted into the connected portions **34** and **44** in the longitudinal direction of the handlebar **4**. The connecting shaft portion **53** can prevent the two clamping members **30** and **40** from being separated. Therefore, for example, the work of attaching the clamping device **100** to the handlebar **4** can be facilitated.

[0079] The first clamping member **30** and the second clamping member **40** can move around the connecting shaft portion **53** so that the distance W2 between the first adjusting portion **33** and the second adjusting portion **43** can be increased or decreased. The clamping members **30** and **40** may be openable so that the distance W2 of the adjusting portions **33** and **43** is greater than a diameter of the outer circumferential surface of the handlebar **4**. Accordingly, it is possible to further facilitate the work of attaching the clamping device **100** to the handlebar **4**.

[0080] A hole **34a** is formed through the first connected portion **34**. The connecting shaft portion **53** is inserted into the hole **34a**. Both ends of the connecting shaft portion **53** protrude from the first connected portion **34** in the longitudinal direction of the handlebar **4**. A connecting hole **44a** is formed through the side wall portion **44c** of the second connected portion **44**. The connecting shaft portion **53** is inserted into the connecting hole **44a**.

[0081] A relative position of the first connected portion **34** and the connecting shaft portion **53** is fixed. In other words, the connecting shaft portion **53** is held by the first connected portion **34**. The connecting shaft portion **53** may be press-fitted into the hole **34a** of the first connected portion **34**. A size of the connecting hole **44a** of the second connected portion **44** is larger than a size of the connecting shaft portion **53**. More specifically, a width of the connecting hole **44a** in a direction extending along a tangent line to the outer circumferential surface of the handlebar **4** (a tangent line passing through the upper end of the outer circumferential surface) and passing through the engaging portions **32** and **42** (a direction extending along the plane P1 in FIG. **6A**) is larger than that of the connecting shaft portion **53**.

[0082] As a result, the connecting shaft portion **53** connects the connected portions **34** and **44** and allows the distance W1 between the first engaging portion **32** and the second engaging portion **42** to change. In an example illustrated in FIG. **5** and the like, the connecting hole **44a** is an elongated hole extending in a direction passing through the engaging portions **32** and **42**. However, the connecting hole **44a** may be circular or elliptical.

[0083] The connecting shaft portion **53** is a member formed separately from the first clamping member **30**, and may be inserted into the holes **34a** and **44a** of the two connected portions **34** and **44** after the two connected portions **34** and **44** are assembled together.

[Position of Connecting Shaft Portion]

[0084] In FIG. **6A**, the plane P1 is a plane extending along a tangent line to the outer circumferential surface of the handlebar **4** (a plane passing through the upper end of the outer circumferential surface) and passing through the two engaging portions **32** and **42**. More specifically, the plane P1 is a plane passing through the protrusion portions **32a** and **42a** of the engaging portions **32** and **42**. In FIG. **6A**, a plane P3 is a plane parallel to the plane P1 and passing through the connecting shaft portion **53**.

[0085] As illustrated in FIG. **6A**, a distance L3 from the outer circumferential surface of the handlebar **4** to the plane P3 is smaller than a distance L1 from the outer circumferential surface of the handlebar **4** to the plane P1. According such an arrangement of the connecting shaft portion **53**, the distance between the outer circumferential surface of the handlebar **4** and the connecting shaft portion **53** is small, so that a change in the relative position of the connecting shaft portion **53** and the connecting hole **44a** of the second connected portion **44** can be decreased. As a result, the connecting hole **44a** having a short length is sufficient, making it easy to ensure the strength of the

second connected portion **44**.

[Positional Relationship Between Engaging Portion and Adjusting Portion]

[0086] In FIG. **6A**, a plane P2 is a plane extending along a tangent line to the outer circumferential surface of the handlebar **4** (a tangent line passing through the lower end of the outer circumferential surface) and passing through the two adjusting portions **33** and **43**. More specifically, the plane P2 is a plane passing through centers of the adjusting portions **33** and **43** (centers of the bolt **51** and the nut **52**).

[0087] The distance L1 from the outer circumferential surface of the handlebar **4** to the plane P1 is greater than a distance L2 from the outer circumferential surface of the handlebar **4** to the plane P2. According to such a structure, the position of the electronic device **70** is far from the handlebar **4**, so that it is easy to avoid interference between the electronic device **70** and other parts attached to the handlebar **4**. When the bolt **51** is strongly tightened, it is possible to prevent an excessive force from acting between the engaging portions **32** and **42** of the clamping members **30** and **40** and the engaged portions **72A** and **72B** of the electronic device **70**. As a result, for example, plastic deformation of the clamping members **30** and **40** can be prevented. Note that the distance L1 from the outer circumferential surface of the handlebar **4** to the plane P1 may be smaller than the distance L2 from the outer circumferential surface of the handlebar **4** to the plane P2, unlike the example illustrated in FIG. **6A**.

[0088] As illustrated in FIG. **6A**, the distance W1 between the first engaging portion **32** and the second engaging portion **42** is greater than the distance W2 between the first adjusting portion **33** and the second adjusting portion **43**. It is preferable that the distance W1 between the first engaging portion **32** and the second engaging portion **42** be greater than the diameter of the outer circumferential surface of the handlebar **4**. Accordingly, the distance W1 becomes large, and support stability of the electronic device **70** by the clamping device **100** can be improved. Meanwhile, it is preferable that the distance W2 between the first adjusting portion **33** and the second adjusting portion **43** be smaller than the diameter of the outer circumferential surface of the handlebar **4**. Accordingly, the fastening unit (the bolt **51** and the nut **52**) can be prevented from standing out due to the handlebar **4**.

[0089] As illustrated in FIG. **6A**, a line B1 is a line passing through an axis Ax of the handlebar **4** and an intermediate position between the first engaging portion **32** and the second engaging portion **42**. When an angular position of the line B1 in the circumferential direction of the handlebar **4** is set to 0 degrees, it is preferable that at least one of the two adjusting portions **33** and **43** is located in a range of 90 degrees to 270 degrees. In other words, it is preferable that at least one of the two adjusting portions **33** and **43** is located between a line B2 (a line indicating a position of 90 degrees) and a line B3 (a line indicating a position of 270 degrees) in the circumferential direction of the handlebar **4**.

[0090] As illustrated in FIG. **6A**, it is preferable that at least one of the two adjusting portions **33** and **43** is located in a range of 135 degrees to 225 degrees. In other words, it is preferable that at least one of the two adjusting portions **33** and **43** is located between a line B4 (a line indicating a position of 135 degrees) and a line B5 (a line indicating a position of 225 degrees) in the circumferential direction of the handlebar **4**. According to such a structure, the two clamping members **30** and **40** can have sizes similar to each other. As a result, a difference in stress acting on the two clamping members **30** and **40** can be decreased.

[0091] In the example illustrated in FIG. **6A**, the two adjusting portions **33** and **43** and the fastening unit (the bolt **51** and the nut **52**) are located in the range from 135 degrees (the line B4) to 225 degrees (the line B5). An intermediate position of the adjusting portions **33** and **43** is located on the opposite side to the intermediate position of the engaging portions **32**, **42** with the axis Ax of the handlebar **4** interposed therebetween. As a result, the engaging portions **32** and **42**, to which a large force is applied, are located separately from the hole **33c** for inserting the bolt **51** formed in the first adjusting portion **33** and an internal gap of the second adjusting portion **43** in which the nut **52** is

disposed. Therefore, it is possible to prevent a decrease in the strength of the engaging portions **32** and **42** (rigidity of the resin around the engaging portions **32** and **42**) caused by the hole **33c** and the gap for the nut **52** being located close thereto.

[0092] As illustrated in FIG. **6A**, the above-described plane P1 passing through the first engaging portion **32** and the second engaging portion **42** and a direction in which the first adjusting portion **33** and the second adjusting portion **43** face each other (insertion direction of the bolt **51**) may be substantially parallel. According to such a structure, during a tightening operation of the fastening unit (specifically, the bolt **51**), the bolt **51** can be easily seen so that the tightening operation is facilitated. For example, a head of the bolt **51** is exposed to the rear, which facilitates the tightening operation of the bolt **51**. Here, “substantially parallel” also includes cases where the plane P1 and the direction in which the adjusting portions **33** and **43** face each other are not completely parallel, but are tilted, for example, in a range of 0 degrees to 10 degrees.

[Another Example of Clamping Device]

[0093] FIGS. **7** and **8** are diagrams illustrating a modification example of the clamping device proposed in the present disclosure. The following description will focus on differences between a clamping device **200** illustrated in FIGS. **7** and **8** and the clamping device **100** illustrated in FIGS. **2** to **6B**. Matters that are not described regarding the clamping device **200** may be similar to those of the clamping device **100**.

[0094] As illustrated in FIG. **7**, the clamping device **200** includes a first clamping member **230** and a second clamping member **240**. The first clamping member **230** has a first engaging portion **232**, an inner circumferential surface **231a** formed along the outer circumferential surface of the handlebar **4**, a first adjusting portion **233**, and a first connected portion **234**. The second clamping member **240** has a second engaging portion **242**, an inner circumferential surface **241a** formed along the outer circumferential surface of the handlebar **4**, a second adjusting portion **243**, and a second connected portion **244**.

[0095] As illustrated in FIG. **8**, the inner circumferential surface **231a** of the first clamping member **230** is formed to cover along only approximately $\frac{1}{4}$ of the outer circumferential surface of the handlebar **4**. The first adjusting portion **233** is located below the first engaging portion **232**. In a plan view of the clamping device **200**, a protrusion portion **232a** of the first engaging portion **232** overlaps with the first adjusting portion **233**. Meanwhile, the inner circumferential surface **241a** of the second clamping member **240** surrounds approximately $\frac{3}{4}$ of the outer circumferential surface of the handlebar **4**. The second adjusting portion **243** is located below the first adjusting portion **233**.

[0096] As illustrated in FIG. **8**, the first engaging portion **232** and the second engaging portion **242** are separated from each other in the front-rear direction (the Y1-Y2 direction in FIG. **8**).

Meanwhile, the first adjusting portion **233** and the second adjusting portion **243** face each other in the up-down direction (the Z1-Z2 direction in FIG. **8**), unlike the example illustrated in FIG. **2** or the like. In other words, a plane P1 extending along a tangent line to the outer circumferential surface of the handlebar **4** and passing through the first engaging portion **232** and the second engaging portion **242** intersects with a direction in which the adjusting portions **233** and **243** face each other. More specifically, the plane P1 and the direction in which the adjusting portions **233** and **243** face each other may be substantially perpendicular. “Substantially perpendicular” includes not only a case in which the angle between the plane P1 and the direction in which the adjusting portions **233** and **243** face each other is 90 degrees, but also a case in which the angle is, for example, 85 degrees to 95 degrees.

[0097] The clamping device **200** includes a fastening unit which fixes a distance W2 between the adjusting portions **233** and **243** (see FIG. **8**). The clamping device **200** may include the bolt **51** and the nut **52** as the fastening unit. The nut **52** may be held by the first adjusting portion **233**. A hole into which the bolt **51** is inserted may be formed in the second adjusting portion **243**. The bolt **51** is inserted into the hole from a lower side (a side opposite to the first adjusting portion **233**) of the

second adjusting portion **243** and fastened to the nut **52**. As a result, the distance W2 between the adjusting portions **233** and **243** is fixed. According to such fastening unit, for example, during a tightening operation of the bolt **51**, it is possible to prevent other parts and devices attached to the handlebar **4** from interfering with the fastening operation.

[0098] A line B1 illustrated in FIG. **8** is a line passing through the axis Ax of the handlebar **4** and an intermediate position between the first engaging portion **232** and the second engaging portion **242**, similar to the line B1 illustrated in FIG. **6A**. When an angular position of the line B1 in the circumferential direction of the handlebar **4** is set to 0 degrees, the second adjusting portion **243** of the two adjusting portions **233** and **243** is located in a range of 90 degrees to 270 degrees. That is, the second adjusting portion **243** is located between a line B2 (line indicating a position of 90 degrees) and a line B3 (line indicating a position of 270 degrees) in the circumferential direction of the handlebar **4**. Meanwhile, the first adjusting portion **233** is located in a range exceeding 270 degrees (the line B3).

[0099] As illustrated in FIG. **7**, a recess portion **231c** is formed between the protrusion portion **232a** of the first engaging portion **232** and the first adjusting portion **233**. A protrusion portion of the first engaged portion **72A** of the electronic device **70** is fitted into the recess portion **231c**. The nut **52** is located below the recess portion **231c**. In the first clamping member **30** illustrated in FIG. **6A**, since the first adjusting portion **233** is located below the handlebar **4**, such a recess portion is not necessary.

[0100] Movements of the first clamping member **230** and the second clamping member **240** are similar to the movements of the clamping members **30** and **40** of the clamping device **100** described above.

[0101] That is, when the bolt **51** is tightened, the distance W2 between the adjusting portions **233** and **243** (see FIG. **8**) decreases. Conversely, when the bolt **51** is loosened, the distance W2 between the adjusting portions **233** and **243** is allowed to increase. A distance W1 between the first engaging portion **232** and the second engaging portion **242** (see FIG. **8**) in the tangential direction of the outer circumferential surface of the handlebar **4** is allowed to decrease or increase.

[0102] When the inner circumferential surfaces **231a** and **241a** interpose the handlebar **4** therebetween, the distance W1 between the first engaging portion **232** and the second engaging portion **242** increases due to a decrease in the distance W2 between the adjusting portions **233** and **243**. Accordingly, a contact force between the inner circumferential surfaces **231a** and **241a** and the outer circumferential surface of the handlebar **4** increases. Therefore, a position of the clamping device **200** on the handlebar **4** is fixed. A relative position of the clamping device **200** and the electronic device **70** in the longitudinal direction of the handlebar **4** is fixed.

[0103] Conversely, an increase in the distance W2 between the adjusting portions **233** and **243** allows the distance W1 between the first engaging portion **232** and the second engaging portion **242** to decrease. Accordingly, a contact force between the first engaging portion **232** and the first engaged portion **72A** and a contact force between the second engaging portion **242** and the second engaged portion **72B** is weakened. As a result, the relative position of the clamping device **200** and the electronic device **70** in the longitudinal direction of the handlebar **4** are allowed to be changed. The contact force between the inner circumferential surfaces **231a** and **241a** and the outer circumferential surface of the handlebar **4** also decreases. Therefore, it is also possible to adjust the position of the clamping device **200** in the longitudinal direction of the handlebar **4**.

[0104] The first clamping member **230** and the second clamping member **240** are connected to each other only via the connecting shaft portion **53**, and are not connected to each other in any other portions. Therefore, when the distance W2 between the adjusting portions **233** and **243** decreases or increases, the distance W1 between the engaging portions **232** and **242** accordingly increases or decreases.

Summary

[0105] (1) Clamping devices **100** and **200** proposed in the present disclosure is for attaching an

electronic device **70** including a first engaged portion **72A** and a second engaged portion **72B** to a handlebar **4**. The clamping devices **100** and **200** includes a first clamping member **30** or **230**, having a first inner circumferential surface **31a** or **231a**, extending along a portion of an outer circumferential surface of the handlebar **4**, a first engaging portion **32** or **232** engageable with the first engaged portion **72A** to allow movement of the first engaged portion **72A** in a longitudinal direction of the handlebar **4**, and a first adjusting portion **33** or **233**, a second clamping member **40** or **240** having a second inner circumferential surface **41a** extending along another portion of the outer circumferential surface of the handlebar **4**, a second engaging portion **42** or **242** engageable with the second engaged portion **72B** to allow movement of the second engaged portion **72B** in the longitudinal direction of the handlebar **4**, and a second adjusting portion **43** or **243**, and a fastening unit configured to fix a distance between the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243** in a circumferential direction of the handlebar **4**. The first engaging portion **32** or **232** and the second engaging portion **42** or **242** are located between the first engaged portion **72A** and the second engaged portion **72B** in a tangential direction of the outer circumferential surface of the handlebar **4**. The fastening unit allows adjustment of a distance $W2$ between the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243**. A distance $W1$ between the first engaging portion **32** or **232** and the second engaging portion **42** or **242** increases due to a decrease in the distance $W2$ between the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243**.

[0106] According to the clamping devices **100** and **200**, both a (a) relative position of the clamping device and the handlebar **4**, and a (b) relative position of the clamping device and the electronic device **70** can be adjusted or fixed by operating the fastening unit. Therefore, a work for position adjustment can be facilitated.

[0107] (2) The clamping devices **100** or **200** according to (1) may further include a connecting shaft portion **53**. The first clamping member **30** or **230** and the second clamping member **40** or **240** respectively have a first connected portion **34** or **234** and a second connected portion **44** or **244** located between the first engaging portion **32** or **232** and the second engaging portion **42** or **242** in the tangential direction of the outer circumferential surface of the handlebar **4**. The connecting shaft portion **53** connects the first connected portion **34** or **234** and the second connected portion **44** or **244** to allow an increase and decrease in the distance $W1$ between the first engaging portion **32** or **232** and the second engaging portion **42** or **242**. Such a structure makes it possible to prevent separation of the two clamping members **30** and **40**. As a result, the clamping devices **100** and **200** are easy to handle.

[0108] (3) In the clamping devices **100** or **200** according to (2), the connecting shaft portion **53** protrudes from the first connected portion **34** or **234** in the longitudinal direction of the handlebar **4**. A connecting hole **44a** into which the connecting shaft portion **53** is fitted may be formed in the second connected portion **44** or **244**. The connecting hole **44a** may be larger than the connecting shaft portion **53** in the tangential direction of the outer circumferential surface of the handlebar **4**.

[0109] (4) In the clamping devices **100** and **200** according to (3), a distance $L3$ between the connecting shaft portion **53** and the outer circumferential surface of the handlebar **4** may be smaller than a distance $L1$ between the outer circumferential surface of the handlebar **4** and a plane $P1$ passing through the first engaging portion **32** or **232** and the second engaging portion **42** or **242**, the plane $P1$ extending along a first tangent line to the outer circumferential surface of the handlebar **4**. Such an arrangement of the connecting shaft portion **53** allows a size of the connecting hole **44a** in the tangential direction of the outer circumferential surface of the handlebar **4** to be decreased. As a result, a decrease in strength of the second clamping member **40** or **240** due to the connecting hole **44a** can be prevented.

[0110] (5) In the clamping devices **100** and **200** according to any one of (1) to (4), the first adjusting portion **33** or **233**, a contact surface between the first inner circumferential surface **31a** or **231a** and the outer circumferential surface of the handlebar **4**, and the first engaging portion **32** or

232 respectively function as a point of effort, a fulcrum point, and a point of load. The second adjusting portion **43** or **243**, a contact surface between the second inner circumferential surface **41a** and the outer circumferential surface of the handlebar **4**, and the second engaging portion **42** or **242** respectively function as a point of effort, a fulcrum point, and a point of load.

[0111] (6) In the clamping devices **100** and **200** according to any one of (1) to (5), the distance L1 between the outer circumferential surface of the handlebar **4** and the plane P1 passing through the first engaging portion **32** or **232** and the second engaging portion **42** or **242** may be greater than a distance L2 between the outer circumferential surface of the handlebar **4** and a plane P2 passing through the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243**, the plane P1 extending along the first tangent line to the outer circumferential surface of the handlebar **4**, and the plane P2 extending along a second tangent line to the outer circumferential surface of the handlebar **4**. Accordingly, when positions of the two adjusting portions **33** and **43**, or **233** and **243** are fixed by the fastening unit, it is possible to prevent an excessive force from acting between the first engaging portion **32** or **232** of the first clamping member **30** or **230** and the first engaged portion **72A** of the electronic device **70**, and prevent an excessive force from acting between the second engaging portion **42** or **242** of the second clamping member **40** or **240** and the second engaged portion **72B** of the electronic device **70**.

[0112] (7) In the clamping devices **100** and **200** according to any one of (1) to (6), the distance W1 between the first engaging portion **32** or **232** and the second engaging portion **42** or **242** in a first direction passing through the first engaging portion **32** or **232** and the second engaging portion **42** or **242** may be greater than the distance W2 between the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243** in a second direction passing through the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243**, the first direction extending along the first tangent line to the outer circumferential surface of the handlebar **4**, and the second direction extending along the second tangent line to the outer circumferential surface of the handlebar **4**. Accordingly, the distance W1 between the first engaging portion **32** or **232** and the second engaging portion **42** or **242** increases, thereby improving stability of the clamping devices **100** and **200** in supporting the electronic device **70**. Since the distance W2 between the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243** decreases, the fastening unit (the bolt **51** and the nut **52**) can be prevented from standing out.

[0113] (8) In the clamping devices **100** and **200** according to any one of (1) to (7), when an angular position of an intermediate position between the first engaging portion **32** or **232** and the second engaging portion **42** or **242** is set to 0 degrees in the circumferential direction of the handlebar **4**, at least one of the first adjusting portion **33** or **233** and the second adjusting portion **43** or **243** may be located in a range of 90 degrees to 270 degrees. Accordingly, it is possible to prevent a decrease in strength of the first clamping member **30** or **230** due to an excessive decrease in a distance between the first adjusting portion **33** or **233** and the first engaging portion **32** or **232**. It is possible to prevent a decrease in strength of the second clamping member **40** or **240** due to an excessive decrease in a distance between the second adjusting portion **43** or **243** and the second engaging portion **42** or **242**.

[0114] (9) In the clamping device **100** according to any one of (1) to (8), when an angular position of an intermediate position between the first engaging portion **32** and the second engaging portion **42** is set to 0 degrees in the circumferential direction of the handlebar **4**, an intermediate position between the first adjusting portion **33** and the second adjusting portion **43** may be located in a range of 135 degrees to 225 degrees. Accordingly, the first clamping member **30** and the second clamping member **40** can have sizes similar to each other. As a result, a difference in stress acting on each of the two clamping members **30** and **40** can be reduced.

[0115] (10) In the clamping device **100** according to any one of (1) to (9), the plane P1 passing through the first engaging portion **32** and the second engaging portion **42** may be substantially parallel to a direction in which the first adjusting portion **33** and the second adjusting portion **43**

face each other, the plane P1 extending along the first tangent line to the outer circumferential surface of the handlebar **4**. Accordingly, the fastening unit can be operated from a front side or a rear side of the clamping device **100** when the clamping device **100** is attached to the handlebar **4**. As a result, position adjustment of the two adjusting portions **33** and **43** by the fastening unit can be facilitated.

[0116] (11) In the clamping device **200** according to any one of (1) to (9), the plane P1 passing through the first engaging portion **232** and the second engaging portion **242** may intersect with a direction in which the first adjusting portion **233** and the second adjusting portion **243** face each other, the plane P1 extending along the first tangent line to the outer circumferential surface of the handlebar **4**.

[0117] (12) An electronic device assembly proposed in the present disclosure includes the clamping device **100**, **200** according to any one of (1) to (11) and the electronic device **70**.

[0118] (13) A bicycle **10** proposed in the present disclosure includes the clamping device **100**, **200** according to any one of (1) to (11), the electronic device **70**, and the handlebar **4**.

Others

[0119] The fastening unit is not limited to the bolt **51** and the nut **52** as long as the fastening unit enables fixing and adjustment of the distance W2 between the adjusting portions **33** and **43**. The fastening unit may be a screw, a lever screw (clamp lever), a cam lever, or the like.

[0120] Although the present invention has been illustrated and described herein with reference to embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples may perform similar functions and/or achieve like results. All such equivalent embodiments and examples are within the spirit and scope of the present invention, are contemplated thereby, and are intended to be covered by the following claims.

Claims

1. A clamping device for attaching an electronic device including a first engaged portion and a second engaged portion to a handlebar, the clamping device comprising: a first clamping member having: a first inner circumferential surface extending along a portion of an outer circumferential surface of the handlebar, a first engaging portion engageable with the first engaged portion to allow movement of the first engaged portion in a longitudinal direction of the handlebar, and a first adjusting portion; a second clamping member having: a second inner circumferential surface extending along another portion of the outer circumferential surface of the handlebar, a second engaging portion engageable with the second engaged portion to allow movement of the second engaged portion in the longitudinal direction of the handlebar, and a second adjusting portion; and a fastening unit configured to fix a distance between the first adjusting portion and the second adjusting portion in a circumferential direction of the handlebar, wherein the first engaging portion and the second engaging portion are located between the first engaged portion and the second engaged portion in a tangential direction of the outer circumferential surface of the handlebar, the fastening unit allows adjustment of the distance between the first adjusting portion and the second adjusting portion, and a distance between the first engaging portion and the second engaging portion increases due to a decrease in the distance between the first adjusting portion and the second adjusting portion.

2. The clamping device according to claim 1, further comprising: a connecting shaft portion, wherein the first clamping member and the second clamping member respectively have a first connected portion and a second connected portion located between the first engaging portion and the second engaging portion in the tangential direction of the outer circumferential surface of the handlebar, and the connecting shaft portion connects the first connected portion and the second connected portion to allow an increase and decrease in the distance between the first engaging

portion and the second engaging portion.

3. The clamping device according to claim 2, wherein the connecting shaft portion protrudes from the first connected portion in the longitudinal direction of the handlebar, a connecting hole into which the connecting shaft portion is fitted is formed in the second connected portion, and the connecting hole is larger than the connecting shaft portion in the tangential direction of the outer circumferential surface of the handlebar.

4. The clamping device according to claim 3, wherein a distance between the connecting shaft portion and the outer circumferential surface of the handlebar is smaller than a distance between the outer circumferential surface of the handlebar and a plane passing through the first engaging portion and the second engaging portion, the plane extending along a tangent line to the outer circumferential surface of the handlebar.

5. The clamping device according to claim 1, wherein the first adjusting portion, a contact surface between the first inner circumferential surface and the outer circumferential surface of the handlebar, and the first engaging portion respectively function as a point of effort, a fulcrum point, and a point of load, and the second adjusting portion, a contact surface between the second inner circumferential surface and the outer circumferential surface of the handlebar, and the second engaging portion respectively function as a point of effort, a fulcrum point, and a point of load.

6. The clamping device according to claim 1, wherein a distance between the outer circumferential surface of the handlebar and a first plane passing through the first engaging portion and the second engaging portion is greater than a distance between the outer circumferential surface of the handlebar and a second plane passing through the first adjusting portion and the second adjusting portion, the first plane extending along a tangent line to the outer circumferential surface of the handlebar, and the second plane extending along a tangent line to the outer circumferential surface of the handlebar.

7. The clamping device according to claim 1, wherein a distance between the first engaging portion and the second engaging portion in a first direction passing through the first engaging portion and the second engaging portion is greater than a distance between the first adjusting portion and the second adjusting portion in a second direction passing through the first adjusting portion and the second adjusting portion, the first direction extending along a tangent line to the outer circumferential surface of the handlebar, and the second direction extending along a tangent line to the outer circumferential surface of the handlebar.

8. The clamping device according to claim 1, wherein when an angular position of an intermediate position between the first engaging portion and the second engaging portion is set to 0 degrees in the circumferential direction of the handlebar, at least one of the first adjusting portion and the second adjusting portion is located in a range of 90 degrees to 270 degrees.

9. The clamping device according to claim 1, wherein when an angular position of an intermediate position between the first engaging portion and the second engaging portion is set to 0 degrees in the circumferential direction of the handlebar, an intermediate position between the first adjusting portion and the second adjusting portion is located in a range of 135 degrees to 225 degrees.

10. The clamping device according to claim 1, wherein a plane passing through the first engaging portion and the second engaging portion is substantially parallel to a direction in which the first adjusting portion and the second adjusting portion face each other, the plane extending along a tangent line to the outer circumferential surface of the handlebar.

11. The clamping device according to claim 1, wherein a plane passing through the first engaging portion and the second engaging portion intersects with a direction in which the first adjusting portion and the second adjusting portion face each other, the plane extending along a tangent line to the outer circumferential surface of the handlebar.

12. An electronic device assembly comprising: a clamping device including first clamping member having: a first inner circumferential surface extending along a portion of an outer circumferential surface of a handlebar, a first engaging portion engageable with a first engaged portion to allow

movement of the first engaged portion in a longitudinal direction of the handlebar, and a first adjusting portion; a second clamping member having: a second inner circumferential surface extending along another portion of the outer circumferential surface of the handlebar, a second engaging portion engageable with a second engaged portion to allow movement of the second engaged portion in the longitudinal direction of the handlebar, and a second adjusting portion; and a fastening unit configured to fix a distance between the first adjusting portion and the second adjusting portion in a circumferential direction of the handlebar, wherein the first engaging portion and the second engaging portion are located between the first engaged portion and the second engaged portion in a tangential direction of the outer circumferential surface of the handlebar, the fastening unit allows adjustment of the distance between the first adjusting portion and the second adjusting portion, and a distance between the first engaging portion and the second engaging portion increases due to a decrease in the distance between the first adjusting portion and the second adjusting portion; and an electronic device configured to be held by the clamping device.

13. A bicycle comprising: a clamping device including first clamping member having: a first inner circumferential surface extending along a portion of an outer circumferential surface of a handlebar, a first engaging portion engageable with a first engaged portion to allow movement of the first engaged portion in a longitudinal direction of the handlebar, and a first adjusting portion; a second clamping member having: a second inner circumferential surface extending along another portion of the outer circumferential surface of the handlebar, a second engaging portion engageable with a second engaged portion to allow movement of the second engaged portion in the longitudinal direction of the handlebar, and a second adjusting portion; and a fastening unit configured to fix a distance between the first adjusting portion and the second adjusting portion in a circumferential direction of the handlebar, wherein the first engaging portion and the second engaging portion are located between the first engaged portion and the second engaged portion in a tangential direction of the outer circumferential surface of the handlebar, the fastening unit allows adjustment of the distance between the first adjusting portion and the second adjusting portion, and a distance between the first engaging portion and the second engaging portion increases due to a decrease in the distance between the first adjusting portion and the second adjusting portion; the bicycle further comprising: an electronic device held by the clamping device; and the handlebar.
